Energies of dilute Fermi gasses

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Many-body quantum system with N particles in large box $\Lambda = [0, L]^3$ and Hamiltonian

$$H_N = \sum_{j=1}^{N} -\Delta_{x_j} + \sum_{1 \le i < j \le N} v(x_i - x_j).$$

- Kinetic energy: Laplacian $-\Delta_{x_j}$
- Interaction energy: Pairwise interaction $v \geq 0$, radial and of compact support.

For ground state energy:

- Spin-$\frac{1}{2}$ Fermi gas: Realize H_n on $L^2_a(\Lambda^n; \mathbb{C}^2) = \bigwedge^n L^2(\Lambda; \mathbb{C}^2)$ or $L^2_a(\Lambda^{n\uparrow}) \otimes L^2_a(\Lambda^{n\downarrow})$ if fixed number of spin $\uparrow$ and $\downarrow$
- Spin-polarized Fermi gas: Realize H_n on $L^2_a(\Lambda^n; \mathbb{C}) = \bigwedge^n L^2(\Lambda; \mathbb{C})$

For the pressure: Realize on Fock spaces

$$\mathcal{F} = \bigoplus_{n=0}^{\infty} \bigwedge^n L^2(\Lambda; \mathbb{C}^2) \quad = \bigoplus_{n=0}^{\infty} L^2_a(\Lambda^n; \mathbb{C}^2) \quad \text{spin-1/2 fermions}$$

$$\mathcal{F} = \bigoplus_{n=0}^{\infty} \bigwedge^n L^2(\Lambda; \mathbb{C}) \quad = \bigoplus_{n=0}^{\infty} L^2_a(\Lambda^n; \mathbb{C}) \quad \text{spin-polarized fermions}$$

Interested in ground state energy in the thermodynamic limit

$$e(\rho) = \lim_{\substack{L\to\infty \\ N/|\Lambda|\to\rho}} \inf_{\substack{\psi\in L^2_a(\Lambda^N) \\ \|\psi\|_{L^2}=1}} \frac{\langle\psi|H_N|\psi\rangle}{|\Lambda|},$$

$$e(\rho_\uparrow,\rho_\downarrow) = \lim_{\substack{L\to\infty \\ N_\sigma/|\Lambda|\to\rho_\sigma}} \inf_{\substack{\psi\in L^2_a(\Lambda^{N_\uparrow})\otimes L^2_a(\Lambda^{N_\downarrow}) \\ \|\psi\|_{L^2}=1}} \frac{\langle\psi|H_N|\psi\rangle}{|\Lambda|}$$

or pressure in the thermodynamic limit

$$\psi(\beta,\mu) = \lim_{L\to\infty} \frac{1}{\beta|\Lambda|} \log \mathrm{Tr}_{\mathcal{F}}\left[e^{-\beta(\mathcal{H}-\mu\mathcal{N})}\right] = \lim_{L\to\infty} \sup_{\Gamma} P_\Lambda[\Gamma]$$

at temperature $T = 1/\beta > 0$ and chemical potential μ.

- $\mathcal{H} = \bigoplus_{n=0}^{\infty} H_n$ is the Hamiltonian on Fock space.
- $\mathcal{N} = \bigoplus_{n=0}^{\infty} n$ is the number operator.
- $P_\Lambda[\Gamma]$ is the pressure functional of the state Γ. A state Γ is an operator on $\mathcal{F}$ with $0 \le \Gamma \le 1$ and $\mathrm{Tr}_{\mathcal{F}}\,\Gamma = 1$.

$$L^3 P_\Lambda[\Gamma] = \mathrm{Tr}_{\mathcal{F}}[\mathcal{H}\Gamma] - \frac{1}{\beta}S(\Gamma) = \mathrm{Tr}_{\mathcal{F}}[\mathcal{H}\Gamma] + \frac{1}{\beta}\,\mathrm{Tr}_{\mathcal{F}}\left[\Gamma\log\Gamma\right].$$

Scattering lengths

Dilute limit: ρ small compared to scattering length of v, i.e. $a^3\rho \ll 1$.

- $\rightsquigarrow$ Expect only pairwise interactions (to leading order)
- $\rightsquigarrow$ Study minimal energy of two particles in a box with
  - no symmetry (fermions of different spin) $\rightsquigarrow$ s-wave: $E \simeq \frac{8\pi a_s}{L^3}$
  - fermionic symmetry (fermions of same spin) $\rightsquigarrow$ p-wave: $E \simeq \frac{4\pi^2}{L^2} + \frac{16\pi^3 a_p^3}{L^5}$

Definition

The s- and p-wave scattering lengths a_s and a_p are defined by

$$4\pi a_s = \inf\left\{\int \left(|\nabla f|^2 + \frac{1}{2}vf^2\right) \mathrm{d}x : f(x) \to 1 \text{ for } |x| \to \infty\right\},$$

$$12\pi a_p^3 = \inf\left\{\int \left(|\nabla f|^2 + \frac{1}{2}vf^2\right) |x|^2 \,\mathrm{d}x : f(x) \to 1 \text{ for } |x| \to \infty\right\}.$$

$$8\pi a_s = \int f_s v \approx \int v, \qquad 24\pi a_p^3 = \int |x|^2 f_p v \approx \int |x|^2 v \qquad \text{for smooth and small } v.$$

Bosons

Realize $H_N = \sum_{j=1}^N -\Delta_{x_j} + \sum_{1\le i<j\le N} v(x_i - x_j)$ on $L^2_s(\Lambda^N) = L^2(\Lambda)^{\otimes_{\mathrm{sym}} N}$.

Theorem (Dyson 1957; Lieb–Yngvason 1998 (+ many others))

For appropriate v we have

$$e_{\mathrm{Bose}}(\rho) = 4\pi a_s \rho^2 \left[1 + \frac{128}{15\sqrt{\pi}}(a_s^3\rho)^{1/2} + o((a_s^3\rho)^{1/2})\right], \qquad a_s^3\rho \ll 1.$$

- Energy of one pair of particles is $8\pi a_s/L^3$. (Short calculation)
- There are $N(N-1)/2 \simeq N^2/2$ many pairs.

$\rightsquigarrow$ Energy density $4\pi a_s \rho^2$.

Goal for fermions:

Find "analogue of $4\pi a_s \rho^2$" for both zero and positive temperature.

Spin-1/2 Fermi gas

Theorem (Lieb–Seiringer–Solovej 2005)

Let $v \geq 0$ be radial and of compact support. Then for $a_s^3\rho \ll 1$

$$e(\rho_\uparrow, \rho_\downarrow) = \frac{3}{5}(6\pi^2)^{2/3}\left(\rho_\uparrow^{5/3} + \rho_\downarrow^{5/3}\right) + 8\pi a_s \rho_\uparrow \rho_\downarrow + a_s \rho^2 \delta(a_s^3\rho),$$

with $\rho = \rho_\uparrow + \rho_\downarrow$ and $-C(a_s^3\rho)^{1/39} \leq \delta(a_s^3\rho) \leq C(a_s^3\rho)^{2/27}$.

Theorem (Seiringer 2006)

Let $v \geq 0$ be radial and of compact support. Then for small $a_s^3\rho$ uniformly in $T = 1/\beta \lesssim \rho^{2/3}$

$$\psi(\beta, \mu) = \psi_0(\beta, \mu) - 2\pi a_s \rho^2 [1 + o(1)].$$

- Kinetic energy/pressure of free gas.
- Interaction energy. Similar to $4\pi a_s \rho^2$ for the Bose gas:
  - Energy of one pair of spin-$\uparrow$ and spin-$\downarrow$ is $8\pi a_s/L^3$. (Short calculation)
  - $N_\uparrow N_\downarrow$ many such pairs $\rightsquigarrow$ Energy density $8\pi a_s \rho_\uparrow \rho_\downarrow$.
- $T_F \sim \rho^{2/3}$ is the **Fermi temperature**. For higher temperatures expect thermal effects stronger than quantum and so gas should behave like classical gas.

Next order

$$e(\rho_\uparrow, \rho_\downarrow) = \frac{3}{5}(6\pi^2)^{2/3}\left(\rho_\uparrow^{5/3} + \rho_\downarrow^{5/3}\right) + 8\pi a_s \rho_\uparrow \rho_\downarrow + a_s \rho^2 \delta(a_s^3 \rho).$$

Expect the **Huang-Yang term** for $\rho_\uparrow = \rho_\downarrow = \rho/2$

$$\delta(a_s^3\rho) = c_{\mathrm{HY}} a_s \rho^{1/3} + o(a_s \rho^{1/3}), \qquad c_{\mathrm{HY}} = \frac{4(11 - 2\log 2)}{35\pi^2}\left(\frac{3}{4\pi}\right)^{4/3}$$

$\rightsquigarrow$ "Optimal" error is $\delta(a_s^3\rho) = O((a_s^3\rho)^{1/3})$.

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$\rightsquigarrow$ "Optimal" error is $\delta(a_s^3\rho) = O((a_s^3\rho)^{1/3})$.

Bounds on the error: (Assume always $v \geq 0$ radial and compactly supported.)

$-(a_s^3\rho)^{1/39} \lesssim \delta(a_s^3\rho) \lesssim (a_s^3\rho)^{2/27}$	(Lieb–Seiringer–Solovej 2005)
$-(a_s^3\rho)^{1/9} \lesssim \delta(a_s^3\rho) \lesssim (a_s^3\rho)^{2/9}$	v smooth (Falconi–Giacomelli–Hainzl–Porta 2021)
$-(a_s^3\rho)^{1/5} \lesssim \delta(a_s^3\rho) \lesssim (a_s^3\rho)^{1/3}$	v smooth (Giacomelli 2023)
$\delta(a_s^3\rho) \lesssim_\varepsilon (a_s^3\rho)^{1/3-\varepsilon}$ for any $\varepsilon > 0$	(L. 2023)

Spin-polarized Fermi gas

Theorem (L.–Seiringer 2024)

Let $v \geq 0$ be radial and of compact support. Then for $a_p^3 \rho \ll 1$

$$e(\rho) \leq \frac{3}{5}(6\pi^2)^{2/3}\rho^{5/3} + \frac{12\pi}{5}(6\pi)^{2/3}a_p^3\rho^{8/3}\left[1 - \frac{1}{14}(6\pi^2)^{2/3}a_p^3 R_{\text{eff}}^{-1}\rho^{2/3}(1+o(1))\right].$$

$$R_{\text{eff}}^{-1} = \frac{1}{10\pi a_p^6}\int_{\mathbb{R}^3}|x|^4\left(|\nabla f_p(x)|^2 + \frac{1}{2}v(x)|f_p(x)|^2\right)\mathrm{d}x \qquad \text{the effective range.}$$

Theorem (L.–Seiringer 2024)

Let $v \in L^1$ be non-negative, radial and of compact support. Then for $a_p^3 \rho \ll 1$

$$e(\rho) \geq \frac{3}{5}(6\pi^2)^{2/3}\rho^{5/3} + \frac{12\pi}{5}(6\pi)^{2/3}a_p^3\rho^{8/3}(1+o(1)).$$

Expected from physics literature with $=$ (Efimov–Amus'ya 1965; Efimov 1966; Ding–Zhang 2019)

- Kinetic energy/energy of free gas.
- Interaction energy.

Theorem (L.–Seiringer 2023)

Let $v \geq 0$ be radial and of compact support. Then for $a_p^3 \rho \ll 1$ uniformly in $T = 1/\beta \lesssim \rho^{2/3}$

$$\psi(\beta, \mu) \geq \psi_0(\beta, \mu) - 12\pi c_{\beta,\mu} a_p^3 \rho^{8/3}[1 + o(1)].$$

$T_F \sim \rho^{2/3}$ is the **Fermi temperature**. For higher temperatures expect thermal effects stronger than quantum and so gas should behave like classical gas.

- Pressure of free gas.
- Interaction energy.

$$c_{\beta,\mu} = 2\pi \frac{-\operatorname{Li}_{5/2}(-e^{\beta\mu})}{(-\operatorname{Li}_{3/2}(-e^{\beta\mu}))^{5/3}}, \qquad -\operatorname{Li}_s(-e^x) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{t^{s-1}}{e^{t-x}+1}\, \mathrm{d}t$$

Depends on temperature — spin-$\frac{1}{2}$ didn't.

1st order perturbation theory — Naive $a^3\rho^{8/3}$

Zero temperature:

- Take trial state ψ_F ground state of free gas:

$$\lim_{\substack{N,|\Lambda|\to\infty \\ N/|\Lambda|=\rho}} \frac{\langle\psi_F|H_N|\psi_F\rangle}{|\Lambda|} = \frac{3}{5}(6\pi^2)^{2/3}\rho^{5/3} + \frac{1}{2}\int \rho^{(2)}(x,0)v(x)\,\mathrm{d}x.$$

- Taylor expand $\rho^{(2)}(x,0) \simeq \frac{(6\pi^2)^{2/3}}{5}\rho^{8/3}|x|^2$ and use $\int |x|^2 v(x)\,\mathrm{d}x \approx 24\pi a_p^3$

$\rightsquigarrow$ Energy density $\frac{3}{5}(6\pi^2)^{2/3}\rho^{5/3} + \frac{12\pi}{5}(6\pi^2)^{2/3}a_p^3\rho^{8/3}$.

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Positive temperature:

- Compute pressure functional of free state:

$$\lim_{L\to\infty} P_\Lambda[\Gamma_F] = \psi_0(\beta,\mu) - \frac{1}{2}\int \rho^{(2)}(x,0)v(x)\,\mathrm{d}x.$$

- Taylor expand $\rho^{(2)}(x,0) \simeq c_{\beta,\mu}|x|^2\rho^{8/3}$ and use $\int |x|^2 v(x) \approx 24\pi a_p^3$.

$\rightsquigarrow$ Pressure $\psi_0 - 12\pi c_{\beta,\mu}a_p^3\rho^{8/3}$. Explains temperature dependence.

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$\rightsquigarrow$ Pressure $\psi_0 - 12\pi c_{\beta,\mu}a_p^3\rho^{8/3}$. Explains temperature dependence.

Need to treat interactions more precisely

$\rightsquigarrow$ 2nd order perturbation theory. (Lower bound on $e(\rho)$.)

$\rightsquigarrow$ Include the scattering functions f_p in trial state. (Upper bound on $e(\rho)$.)

2nd order perturbation theory

$$H_N = H^0 + V, \qquad H^0 = \sum_j -\Delta_j, \qquad V = \sum_{i<j} v_{ij} = \sum_{i<j} v(x_i - x_j).$$

Treat only "off-diagonal" part of V as the perturbation:

$$V = \sum_{i<j} v_{ij} \approx \underbrace{\sum_{i<j} P_i P_j v_{ij} P_i P_j + Q_i Q_j v_{ij} Q_i Q_j}_{V_{\mathrm{D}}:=} + \underbrace{\sum_{i<j} Q_i Q_j v_{ij} P_i P_j + P_i P_j v_{ij} Q_i Q_j}_{V_{\mathrm{OD}}:=}.$$

P and Q projections inside and outside Fermi ball.

Ground state is $\approx e^B \psi_F$ for appropriate unitary e^B.

$$e^{-B} H_N e^B \approx H^0 + \Big(V_{\mathrm{D}} + V_{\mathrm{OD}} + [H^0 + V_{\mathrm{D}}, B] \Big) + \Big([V_{\mathrm{OD}}, B] + \frac{1}{2}[[H^0 + V_{\mathrm{D}}, B], B] \Big).$$

2nd order perturbation theory

$$H_N = H^0 + V, \qquad H^0 = \sum_j -\Delta_j, \qquad V = \sum_{i<j} v_{ij} = \sum_{i<j} v(x_i - x_j).$$

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P and Q projections inside and outside Fermi ball.

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$$e^{-B} H_N e^B \approx H^0 + \Big(V_{\mathrm{D}} + V_{\mathrm{OD}} + [H^0 + V_{\mathrm{D}}, B]\Big) + \Big([V_{\mathrm{OD}}, B] + \frac{1}{2}[[H^0 + V_{\mathrm{D}}, B], B]\Big).$$

B chosen to cancel off-diagonal part: $[H^0 + V_{\mathrm{D}}, B] \approx -V_{\mathrm{OD}}$. Achieved with (by neglecting 3-body terms and writing $Q \approx \mathbb{1}$)

$$B = \sum_{i<j} P_i P_j (1 - f_p)_{ij} Q_i Q_j - Q_i Q_j (1 - f_p)_{ij} P_i P_j$$

since f_p solves *p-wave scattering equation* (Euler–Lagrange equation of minimization problem)

$$2|x|^2 \Delta f_p + 4x \cdot \nabla f_p - |x|^2 v f_p = 0.$$

2nd order perturbation theory

Conclude

$$\inf_{\|\psi\|=1} \langle \psi | H_N | \psi \rangle \approx \left\langle \psi_F \middle| H^0 \middle| \psi_F \right\rangle + \langle \psi_F | V_{\mathrm{D}} | \psi_F \rangle + \frac{1}{2} \langle \psi_F | [V_{\mathrm{OD}}, B] | \psi_F \rangle .$$

with

$$V_{\mathrm{D}} = \sum_{i<j} P_i P_j v_{ij} P_i P_j + Q_i Q_j v_{ij} Q_i Q_j, \qquad V_{\mathrm{OD}} = \sum_{i<j} Q_i Q_j v_{ij} P_i P_j + P_i P_j v_{ij} Q_i Q_j,$$

$$B = \sum_{i<j} P_i P_j (1 - f_p)_{ij} Q_i Q_j - Q_i Q_j (1 - f_p)_{ij} P_i P_j.$$

2nd order perturbation theory

Conclude

$$\inf_{\|\psi\|=1} \langle\psi|H_N|\psi\rangle \approx \left\langle \psi_F \middle| H^0 \middle| \psi_F \right\rangle + \langle\psi_F|V_{\mathrm{D}}|\psi_F\rangle + \frac{1}{2}\langle\psi_F|[V_{\mathrm{OD}}, B]|\psi_F\rangle\,.$$

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$$B = \sum_{i<j} P_i P_j (1-f_p)_{ij} Q_i Q_j - Q_i Q_j (1-f_p)_{ij} P_i P_j.$$

Compute $[V_{\mathrm{OD}}, B]$ by neglecting 3-body terms and writing $Q \approx \mathbb{1}$:

$$[V_{\mathrm{OD}}, B] \approx -2 \sum_{i<j} P_i P_j v_{ij} (1-f_p)_{ij} P_i P_j$$

Conclude

$$e(\rho) \approx \frac{3}{5}(6\pi^2)^{2/3}\rho^{5/3} + \int \rho^{(2)}(x,0)v(x)\,\mathrm{d}x - \int \rho^{(2)}(x,0)v(x)(1-f_p(x))\,\mathrm{d}x$$

Taylor $\rho^{(2)}(x,0) \simeq \frac{(6\pi^2)^{2/3}}{5}\rho^{8/3}|x|^2$ and use $\int |x|^2 f_p v = 24\pi a_p^3$. Conclude

$$e(\rho) \approx \frac{3}{5}(6\pi^2)^{2/3}\rho^{5/3} + \frac{12\pi}{5}(6\pi^2)^{2/3} a_p^3 \rho^{8/3}.$$

Cluster expansion

Jastrow-type trial state

$$\psi_N(x_1, \ldots, x_N) = \frac{1}{\sqrt{C_N}} \prod_{1 \le i < j \le N} f_p(x_i - x_j) D_N(x_1, \ldots, x_N)$$

D_N Slater determinant, C_N normalization constant.

Compute energy

$$\langle \psi_N | H_N | \psi_N \rangle = E_0 + \iint \rho^{(2)}_{\text{Jas}}(x_1, x_2) \left(\left| \frac{\nabla f_p(x_1 - x_2)}{f_p(x_1 - x_2)} \right|^2 + \frac{1}{2} v(x_1 - x_2) \right) \mathrm{d}x_1 \, \mathrm{d}x_2$$

$$+ \text{3-body term}$$

E_0 kinetic energy of Slater determinant, $\rho^{(2)}_{\text{Jas}}$ the 2-particle density. Think $\rho^{(2)}_{\text{Jas}}(x_1, x_2) \simeq f_p(x_1 - x_2)^2 \rho^{(2)}(x_1, x_2)$. (Recall $12\pi a_p^3 = \int (|\nabla f_p|^2 + \frac{1}{2} v f_p^2)|x|^2 \, \mathrm{d}x$.)

Compute $\rho^{(2)}_{\text{Jas}} \rightsquigarrow$ **Cluster expansion**: Way of calculating $\rho^{(n)}_{\text{Jas}}$. (Formal calculations of Gaudin–Gillespie–Ripka 1971.)

$$\rho^{(2)}_{\text{Jas}}(x_1, x_2) = f(x_1 - x_2)^2 \left[\rho^{(2)}(x_1, x_2) + \sum_{p=1}^{\infty} \frac{1}{p!} \sum_{(\pi, G) \in \tilde{\mathcal{L}}^2_p} \Gamma^2_{\pi, G}(x_1, x_2) \right]$$

$$= f(x_1 - x_2)^2 \left[\rho^{(2)}(x_1, x_2) + \int \rho^{(3)} \left(g_{13} + g_{23} + g_{13} g_{23} \right) \mathrm{d}x_3 + \ldots \right]$$

$g_{ij} = f(x_i - x_j)^2 - 1.$

Explicit description of $\tilde{\mathcal{L}}^2_p$ as a set of diagrams.

Valid for both zero and positive temperature

$$\rho^{(2)}_{\mathrm{Jas}}(x_1,x_2) = f(x_1-x_2)^2 \left[\rho^{(2)}(x_1,x_2) + \sum_{p=1}^{\infty} \frac{1}{p!} \sum_{(\pi,G)\in\tilde{\mathcal{L}}^2_p} \Gamma^2_{\pi,G}(x_1,x_2) \right]$$

$$= f(x_1-x_2)^2 \left[\rho^{(2)}(x_1,x_2) + \int \rho^{(3)} \left(g_{13} + g_{23} + g_{13}g_{23} \right) \mathrm{d}x_3 + \dots \right]$$

$g_{ij} = f(x_i - x_j)^2 - 1.$

Explicit description of $\tilde{\mathcal{L}}^2_p$ as a set of diagrams.

Valid for both zero and positive temperature

Absolute convergence:

$$\sum_{p=1}^{\infty} \frac{1}{p!} \left| \sum_{(\pi,G)\in\tilde{\mathcal{L}}^2_p} \Gamma^2_{\pi,G} \right| < \infty.$$

(Non-trivial) combinatorics: reduces to

$$\rho \int |g(x)| \, \mathrm{d}x \int |\gamma^{(1)}(x;0)| \, \mathrm{d}x = \rho \times a_p^3 \left| \log a_p^3 \rho \right| \times \int_{[0,L]^3} |\gamma^{(1)}(x)| \, \mathrm{d}x < c.$$

$\gamma^{(1)}$ one particle density matrix of Slater determinant D_N.

Absolute convergence at zero temperature

Absolute convergence

$\gamma^{(1)}(x) = \frac{1}{L^3} \sum_{k \in M_F} e^{ikx}$. Thus absolute convergence if

$$a_p^3 \rho \left| \log a_p^3 \rho \right| \times \frac{1}{L^3} \int_{[0,L]^3} \left| \sum_{k \in M_F} e^{ikx} \right| \mathrm{d}x < c.$$

Introduce Fermi polyhedron:

For Fermi ball B_F and Fermi polyhedron P_F (Kolomoitsev–Lomako 2018):

$$\frac{1}{L^3} \int_{[0,L]^3} \left| \sum_{k \in B_F} e^{ikx} \right| \mathrm{d}x \sim N^{1/3}, \qquad \frac{1}{L^3} \int_{[0,L]^3} \left| \sum_{k \in P_F} e^{ikx} \right| \mathrm{d}x \lesssim s(\log N)^3.$$

s is number of corners of P_F – some parameter to choose.

- Finite size error from kinetic energy $\sim N^{-1/3} \rho^{5/3}$.
- Leading energy density from interaction $\sim a_p^3 \rho^{8/3}$.

$\rightsquigarrow$ Need $N^{-1/3} \rho^{5/3} \ll a_p^3 \rho^{8/3}$, i.e. $N^{1/3} a_p^3 \rho \gg 1$.

$\rightsquigarrow$ **Can't use Fermi ball.**

Absolute convergence at positive temperature

Absolute convergence

$$\gamma^{(1)}(x) = \frac{1}{L^3} \sum_{k \in \frac{2\pi}{L}\mathbb{Z}^3} \frac{1}{1 + e^{\beta|k|^2 - \beta\mu}} e^{ikx}, \qquad \int_\Lambda |\gamma^{(1)}(x)| \, \mathrm{d}x \leq C\rho\beta^{3/2}.$$

Thus absolute convergence if

$$a_p^3 \rho \left| \log a_p^3 \rho \right| \beta^{3/2} \rho < c.$$

Uniform in volume

Satisfied for not too small temperatures:

$$T = \frac{1}{\beta} > C\rho^{2/3} (a_p^3 \rho |\log a_p^3 \rho|)^{2/3} \sim T_F (a_p^3 \rho |\log a_p^3 \rho|)^{2/3}.$$

For smaller temperatures compare to zero-temperature problem to prove theorem.

Thanks for your attention!

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